

JANGHO YANG

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PROFESSIONAL SUMMARY

My research focuses on technological change, productivity dynamics, and firm innovation, integrating information theory, Bayesian modeling, and distributional analysis. I hold positions at the University of Waterloo and the University of Oxford.

CURRENT POSITION

Assistant Professor, University of Waterloo *2020 - Present*
Department of Management Sciences, Faculty of Engineering

- *Affiliated Member, Waterloo Data and Artificial Intelligence Institute*
- *Affiliated Member, Waterloo Institute for Complexity and Innovation*

Research Associate, University of Oxford *2020 - Present*
Institute for New Economic Thinking at the Oxford Martin School

PREVIOUS ACADEMIC POSITION

Postdoctoral Research Fellow, University of Oxford *2018 - 2020*
Oxford Martin Programme on Technological and Economic Change
Institute for New Economic Thinking at the Oxford Martin School

EDUCATION

Ph.D. in Economics *2015 - 2018*
New School for Social Research, New York, NY
Dissertation Title: “Essays on the Statistical Equilibrium Model of Technical Change”
Dissertation Committee: Duncan Foley (Chair), Paulo dos Santos, Mark Setterfield, Deepankar Basu

Master’s in Economics *2013 - 2015*
New School for Social Research, New York, NY

Bachelor in Western History *2005 - 2010*
Seoul National University, Seoul, Korea

RESEARCH INTERESTS

Economics of technological change; productivity distribution and firm-level dynamics; information-theoretic approaches in economics; Bayesian multilevel modeling; power-law patterns in economic data.

TEACHING EXPERIENCE

Department of Management Sciences, University of Waterloo: *Quantitative Data Analysis* (2021-2023); *Statistical Methods for Data Analytics* (2022-2025); *Engineering Economics: Financial Management for Engineers* (2021-2023)

Department of Economics, New School for Social Research: *Advanced Econometrics* (TA and lab consultant, Fall 2017, Spring 2017, Spring 2016)

1. Ilan Strauss, **Jangho Yang**, Tim O'Reilly, Sruly Rosenblat, and Isobel Moure (2026), "The Attribution Crisis in LLM Search Results: Estimating Ecosystem Exploitation," *Data & Policy*. DOI: 10.1017/dap.2026.10064
2. Ilan Strauss and **Jangho Yang** (2026), "M&A as Capability Building: Patent Evidence from Big Tech," *Applied Economics Letters*. DOI: 10.1080/13504851.2026.2651435
3. Bonwoo Koo, **Jangho Yang**, and Brian D. Wright (2025), "Protecting Intermediate Innovations When Ideas Are Scarce: Patents or Secrecy?," *Journal of Economics & Management Strategy*. DOI: 10.1111/jems.70001
4. **Jangho Yang**, Brian Cozzarin, and Torsten Heinrich (2025), "Productivity Dispersion and Investment Slowdown in Canada," *Journal of Evolutionary Economics*. DOI: 10.1007/s00191-025-00891-z
5. **Jangho Yang**, Torsten Heinrich, Julian Winkler, François Lafond, Pantelis Koutroumpis, and J. Doyne Farmer (2024), "Measuring Productivity Dispersion: A Parametric Approach Using the Lévy Alpha-Stable Distribution," *Industrial and Corporate Change*. DOI: 10.1093/icc/dtae021
6. Ilan Strauss and **Jangho Yang** (2024), "Testing for secular stagnation in investment rates using a Bayesian multilevel model," *Structural Change and Economic Dynamics*. DOI: 10.1016/j.strueco.2024.03.011
7. **Jangho Yang** (2022), "Information-Theoretic Model of Induced Technological Change: Theory and Empirics," *Metroeconomica*. DOI: 10.1111/meca.12399
8. Torsten Heinrich, **Jangho Yang**, and Shuanping Dai (2021), "Levels of structural change: An analysis of China's development push 1998–2014," *Journal of Evolutionary Economics*. DOI: 10.1007/s00191-021-00740-9
9. Ilan Strauss and **Jangho Yang** (2021), "Slowing investment rates in developing economies: Evidence from a Bayesian hierarchical model," *International Review of Financial Analysis*. DOI: 10.1016/j.irfa.2021.101843
10. **Jangho Yang** and Adrián Carro (2020), "Two Tales of Complex System Analysis: MaxEnt and Agent-Based Modeling," *European Physical Journal Special Topics*. DOI: 10.1140/epjst/e2020-900137-x
11. Ellis Scharfenaker and **Jangho Yang** (2020), "Maximum Entropy Economics," *European Physical Journal Special Topics*. DOI: 10.1140/epjst/e2020-000029-4
12. Paulo dos Santos and **Jangho Yang** (2020), "Capital Mobility, Quasi-rents, and the Competitive Self-organization of Distributions of Profitability," *Advances in Complex Systems*. DOI: 10.1142/S0219525920500034
13. Paulo dos Santos and **Jangho Yang** (2019), "The persistent and informative distribution of returns on capital," *Economics and Business Letters*. DOI: 10.17811/eb1.8.3.2019.156–165
14. **Jangho Yang** and Luis Daniel Torres Gonzales (2019), "The Persistent Statistical Structure of the US Input–Output Coefficient Matrices: 1963–2007," *Economic Systems Research*. DOI: 10.1080/09535314.2018.1561425
15. **Jangho Yang** (2018), "A Quantal Response Statistical Equilibrium Model of Induced Technical Change in an Interactive Factor Market: Firm-Level Evidence in the EU Economies," *Entropy*. DOI: 10.3390/e20030156

16. **Jangho Yang** (2018), “Information-Theoretic Approaches in Economics,” *The Journal of Economic Surveys*. DOI: 10.1111/joes.12226

WORKING PAPERS (UNDER REVIEW AND R&R)

1. **Jangho Yang**, Ilan Strauss, and Mariana Mazzucato, “Rich-Get-Richer? Analyzing Content Creator Earnings Across Large Social Media Platforms,” URL: <https://ssrn.com/abstract=5253032>
2. **Jangho Yang** and Christian Schoder, “Impact of Temperature Uncertainty on Firm Growth: A Grid-Level Analysis,” URL: <https://hdl.handle.net/10986/42644>
3. Sujung Jee, **Jangho Yang**, Torsten Heinrich, and Kejia Zhu, “Flexibility Stigma and Gender in Innovation: Evidence from Inventors During Lockdown.”

RESEARCH GRANTS

New Frontiers in Research Fund - Special Call. \$241,500 CAD. Feb 2023 - Jan 2027 (with one year automatic extension).

PI: **Jangho Yang**. Co-PIs: Su Jung Jee, Torsten Heinrich, Kejia Zhu.

UW-SSHRC Explore Grant. \$7,000 CAD Jan 2023 - Dec 2024 (with one year automatic extension).

PI: **Jangho Yang**.

International Research Partnership Grant, University of Waterloo. \$20,000 CAD. Mar 2022 - Feb 2023.

PI: **Jangho Yang**.

Economic Security Project. \$25,000 USD. Mar 2022 - Feb 2023.

PIs: **Jangho Yang** and Ilan Strauss.

Treasury Board Secretariat, Canada. \$27,000 CAD. Mar 2022 - Feb 2023.

PIs: **Jangho Yang** and Brian Cozzarin.

Starter Grant, University of Waterloo. \$45,000 CAD.

SELECTED CONSULTING AND ADVISORY WORK

Short-Term Consultant, World Bank

2023 – 2024

Conducted empirical analysis of climate-related uncertainty and its effects on firm investment behavior, drawing on firm-level panel data and econometric modeling.

AWARDS

Edith Henry Johnson Memorial Award in Economics, Civil Affairs, And Education *2018*
awarded for the dissertation “Essays on the Statistical Equilibrium Model of Technical Change”

Eugene Lang College Social Science Fellowship

2017-2018

Outstanding Graduate Student Teacher Award

2016

New School for Social Research Prize Fellowship

2015-2018

full-funding for a PhD program

SUPERVISION

Current PhD Students

(Co-supervision indicated by names in parentheses)

Sepideh Khalaj, Management Science and Engineering, Aug 2025 – Present

Jihae Hwang (Stan Dimitrov), Management Science and Engineering, Aug 2024 – Present

Ho Yi Wong (Bonwoo Koo), Management Science and Engineering, Aug 2024 – Present

Julia D'Alessandro (Bonwoo Koo), Management Science and Engineering, Aug 2023 – Present

Wendy de Gomez (Kejia Zhu), Management Science and Engineering, May 2023 – Present

Bolaji Vincent (Stan Dimitrov), Management Science and Engineering, Sep 2021 – Present

Current MASc Students

Yuhan Zhang (Kejia Zhu), Management Science and Engineering, Aug 2024 – Present

Edward Jeong (Kejia Zhu), Management Science and Engineering, August 2025 – Present

Eugene Kayal, Management Science and Engineering, Jan 2025 – Present

Previous PhD Students

Kirsten Wright (Sean Geobey), Systems Design Engineering, Sep 2021 – 2024 (Graduated)

Previous PhD Students

Lida Ghasemi (Stan Dimitrov), Management Science and Engineering, May 2023 – 2025 (Graduated)

CONFERENCE PRESENTATION

2025 Canadian Economic Association, Montreal, Canada

2024 Info-Metric Workshop, Washington D.C., US

2023 Canadian Economic Association, Winnipeg, Canada

2022 Canadian Economic Association, Ottawa, Canada

2019 Nuffield College Postdoctoral Seminar, Oxford, UK

Workshop on Economic Science with Heterogeneous Interacting Agents, London, UK

International Input-Output Conference, Glasgow, UK

Eastern Economic Association Conference, New York, US

2018 European Association for Evolutionary Political Economy, Nice, France

PROFESSIONAL SERVICE

Editorial Service

Guest Editor, European Physical Journal Special Topics

Co-editing the special issue on “Maximum Entropy Economics: Foundations and Applications,” 2021, with Ellis Scharfenaker.

Referee Service

Oxford University Press, Journal of Economic Growth, Cambridge Journal of Economics, OMEGA, Journal of Post Keynesian Economics, Economic Systems Research, European Physical Journal Special Topics, Metroeconomica, International Review of Financial Analysis, Structural Change and Economic

*Dynamics, Journal of Economic Interaction and Coordination, Journal of Evolutionary Economics,
Review of Social Economy, Sustainability, Big Data and Cognitive Computing*